🌱 What's Inside a Seed? 🌱

Life Science — Plants & Seeds · Unit 2, Lesson 1 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🌿 **Subject:** Life Science — Plants & Seeds 🧰 **Kit Lesson:** No

A tiny seed holds a big secret — a baby plant, packed and ready to grow.

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will investigate what's Inside a Seed?, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

A seed is not just a little brown lump! Inside every seed there is a **baby plant** waiting to wake up. The seed also carries its own *food supply* — everything it needs to start growing before it can make food from the sun. Today we're going to crack one open and look inside.

🔬 The Science

Every seed has three main parts:

- **Seed coat** — the tough outer covering that protects everything inside, like a little sleeping bag.
- **Embryo** — the tiny baby plant curled up inside, waiting for the right conditions to sprout.
- **Cotyledon (food storage)** — the built-in food supply that feeds the baby plant while it's too small to make its own food from sunlight.

When a seed gets water, warmth, and the right conditions, the embryo wakes up and starts to grow — using the stored food as fuel. This is called *germination*.

💡 **Key vocabulary:** seed coat, embryo, cotyledon, germination, observe

📦 Materials

🛒 What You'll Need

- Dried lima beans (a small handful — you'll want extras)
- Cup of water (for overnight soaking)
- Paper towels
- Magnifying glass
- Pencil and paper (or the worksheet on the last page)

⏰ Prep Required

- Soak the lima beans in water the **night before** (or at least 4–6 hours ahead). They'll puff up and become soft — much easier for little fingers to split open!

🌿 Let's Get Started

Step 1 — The Before Observation (5 min)
Put one *dry* lima bean and one *soaked* lima bean on a paper towel side by side. Ask:
"What do you notice? How are these two beans different?"
Let the child touch both. The soaked bean should feel softer and look swollen. Discuss: the water got in through tiny openings in the seed coat!

Step 2 — Peel the Seed Coat (5 min)
Take a soaked bean. Gently help your child peel off the thin papery outer layer — this is the **seed coat**! It should slip off easily.
Say: *"This is the seed's protective covering — like a jacket. What do you think is inside?"*

Step 3 — Split It Open! (5 min)
Hold the bean with a thumbnail at the natural seam and gently press. It should split into two halves. (Do this with your child — it's very satisfying!)
"Look very carefully. Can you see something tiny inside?"
Have the child use the **magnifying glass** to look at the inner curve of one half. That tiny curved shape is the **embryo** — the baby plant!

Step 4 — Find All Three Parts (5–10 min)
With the magnifying glass, help the child identify:

- The **seed coat** (the peeled-off skin)
- The **embryo** (the tiny curved baby plant inside — look for a small nub or sprout shape)
- The **cotyledon** (the two big fleshy halves — this is the food storage!)

Point out: the two big halves are what feeds the baby plant when it first starts to grow. It's like a packed lunch!

Step 5 — Draw What You See (10 min)
While the bean halves are still on the paper towel, have the child draw:
1. The outside of the seed (before splitting)
2. The inside of the seed (split open, both halves visible)
Use the worksheet on the last page to label and circle the baby plant part.

Step 6 — Compare & Wonder (5 min)
Put the dry bean back next to the open soaked bean. Ask: *"This same tiny baby plant was in the dry bean all along. What do you think it was waiting for?"*
(Answer: water! And warmth, and a good place to grow.) Let the child's ideas lead the conversation.

💬 Discussion Questions

🗣️ **Talk about these together:**

- What are the three parts of the seed we found today?

  The seed coat (the outside skin), the baby plant (embryo), and the food storage (cotyledon).

- Why do you think the seed has food stored inside it?

  The baby plant needs food to start growing before it can make its own food from the sun. It's like a packed lunch for the journey!

- How was the soaked bean different from the dry bean?

  The soaked bean was softer, bigger, and easier to open because water got inside and woke it up a little.

- If you were a seed, what would you need to start growing?

  Water, warmth, and soil — let the child brainstorm freely!

🚀 Extensions

- 🍎 **Seed Safari:** Collect and compare different kinds of seeds — apple seeds, sunflower seeds, an avocado pit, a peach pit. Which ones can you split open? What do they have in common?
- 💧 **Germination Watch:** Place a soaked lima bean in a zip-lock bag with a damp paper towel and tape it to a sunny window. Check it daily — how many days until you see the first root or sprout?
- 🌻 **Seed Size Compare:** Line up seeds from smallest to largest. A sesame seed vs. a lima bean vs. an avocado pit — do bigger seeds always grow bigger plants?
- 📚 **Read Aloud:** *"The Tiny Seed"* by Eric Carle or *"From Seed to Plant"* by Gail Gibbons

👩‍🏫 Parent/Teacher Notes

- **Prep tip:** Soak the beans the night before (or first thing in the morning if doing this in the afternoon). Dry beans are too hard for small fingers to split cleanly — the soaked ones make a much more satisfying and clear observation.
- **The embryo is tiny!** A magnifying glass is essential here. Look for a small curved nub on the inner edge of one of the cotyledon halves — it may look like a tiny comma or sprout shape. If you can't see it clearly, try a few beans — some show it more clearly than others.
- **Extension idea:** Compare different seed types side by side — apple seeds, sunflower seeds, a bean, and an avocado pit all have the same basic three-part structure, just at very different scales. This is a great conversation starter about how all seeds are related.
- **Vocabulary to introduce:** *seed coat*, *embryo*, *cotyledon*. You don't need to drill these — just use them naturally as you explore. They'll hear them again in later lessons.
- **This is Lesson 1 of 6:** Don't plant the bean yet — that's coming in Lesson 2! For now, keep the focus on *observation*. We're building the question: "What does a seed need to grow?" before we try to answer it.
- **Science talk tip:** When the child says something like "it has a baby inside!" — that's exactly right. Affirm it and layer in the vocabulary: "Yes! Scientists call that tiny baby plant the *embryo*."

🌱 What's Inside a Seed? — Student Worksheet

Name: _____ Date: _____

Draw what you see! Look closely with your magnifying glass. 🔍

🫘 The Outside of the Seed

✂️ The Inside (Split Open)

🏷️ Label the Parts

Draw a line from each word to the part of your seed drawing, or write the name next to the correct part.

🧥 **seed coat** | 🌿 **baby plant** | 🍱 **food storage**

⭕ Circle It!

In your drawing of the inside, **circle the part that will become a new plant.**

✏️ My Observations

How was the soaked bean different from the dry bean?

What surprised you about the inside of the seed?

🌱 ANSWER KEY — Parent/Teacher Copy

🏷️ Labeling the Seed Parts

- **Seed coat** — The thin papery outer skin that wraps the seed. It peels off easily after soaking. Its job is to protect the embryo and food supply inside.
- **Baby plant (embryo)** — The tiny curved shape tucked inside one of the bean halves. It looks like a little sprout or comma shape. This is the part that will grow into a new plant.
- **Food storage (cotyledon)** — The two large, fleshy halves of the bean. These store the nutrients the embryo uses to start growing before it can make its own food from sunlight.

⭕ Circle the Part That Will Become a New Plant

The child should circle the **embryo** (the baby plant) — the small curved structure visible on the inner edge of one of the cotyledon halves.

✏️ Sample Observation Answers

How was the soaked bean different from the dry bean?

- The soaked bean was bigger and softer than the dry bean.
- The soaked bean was easier to open and peel.
- The soaked bean looked swollen because it absorbed water.

What surprised you about the inside of the seed?

- There was a tiny baby plant hiding inside!
- The big parts are actually food for the baby plant.
- It has a protective coat, just like an egg has a shell.

💬 Discussion Answer

"Why do you think the seed has food stored inside it?"

The baby plant needs food to start growing before it can make its own food from the sun. It's like a packed lunch for the journey! The stored food in the cotyledon feeds the embryo until the seedling grows leaves and can begin photosynthesis.

✅ What Success Looks Like

- Child can name the three parts of a seed: seed coat, baby plant (embryo), food storage (cotyledon).
- Child correctly identifies the embryo as the part that will become a new plant.
- Child can describe at least one way the soaked bean looked or felt different from the dry bean.
- Child can explain (in their own words) why the seed stores food inside.

> 💡 **Coming up in Lesson 2:** We'll plant those beans! Now that kids know what's inside a seed, they're ready to ask: what does a seed need to grow?

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🌱 Planting Day: What Do Seeds Need? 🌱

Life Science — Unit 2, Lesson 2 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🌿 **Subject:** Life Science — Plants & Seeds 🧰 **Kit Lesson:** No

A tiny seed has everything it needs to become a plant — we just have to give it the right conditions!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will investigate planting Day: What Do Seeds Need?, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Seeds are like tiny sleeping plants! Inside every seed is everything needed to start a new plant — but it won't wake up without the right conditions. In this lesson, students will plant three cups of seeds differently and **predict** which ones will grow. Over the next week, they'll **observe** what happens and discover what seeds truly need to sprout.

🔬 The Science

The process of a seed beginning to grow is called **germination**. For germination to happen, seeds need three key things:

- 💧 **Water (moisture)** — softens the seed coat and activates growth
- 🌡️ **Warmth** — provides the energy to trigger the growth process
- 💨 **Air** — seeds need oxygen to fuel their early growth

Soil provides nutrients that help a plant grow big and strong *after* germination — but a seed doesn't need soil to start sprouting! That's why seeds can germinate on a wet paper towel. The dry cup won't sprout at all because without water, the seed stays dormant (asleep).

💡 **Key vocabulary:** seed, germinate, sprout, predict, observe, root, soil, moisture, dormant

📦 Materials

🌱 For Each Experiment Setup

- 3–6 dried bean seeds (kidney beans, lima beans, or black beans from the pantry work great!)
- 3 clear plastic cups (clear is best — kids can watch roots develop through the sides!)
- Potting soil
- Paper towels
- Water
- Sunny windowsill
- Markers for labeling cups

✏️ For the Worksheet

- Printed worksheet pages (this lesson, last section)
- Crayons or markers
- Pencil

🔬 Let's Get Started

Step 1 — Introduction: What Does a Seed Need? (5 min)
Hold up a dried bean seed. Ask: *"What is this? What does it have inside it?"* Tell students it's a sleeping plant! Ask: *"What do YOU need when you're sleeping and want to wake up and grow?"* (Food, water, warmth…) Guide kids to think: what might a tiny seed need to wake up?

Step 2 — Set Up the Three Cups (10–15 min)
Label each cup clearly with a marker:

🟢 **Cup 1: "Soil + Water"** — Fill with potting soil, push 1–2 bean seeds about 1 inch deep, water gently.
🟡 **Cup 2: "Soil Only"** — Fill with dry potting soil, push in 1–2 seeds. *Do NOT add any water.*
🔵 **Cup 3: "No Soil + Water"** — Fold a damp paper towel, place it in the bottom of the cup, tuck 1–2 bean seeds inside or against the edge of the cup. Keep the paper towel moist.

Place all three cups in the **same sunny windowsill** so light and temperature are equal for all three.

💡 **Pro tip:** If using clear cups, press the seeds against the inner wall so students can see root and shoot development from the outside — much more exciting than digging around in soil!

Step 3 — Make Predictions (5 min)
Before anything happens, ask the **Discussion Question** (below). Have each student circle YES or NO for each cup on their worksheet. There are no wrong answers right now — that's what science is for! Write down their predictions together.

Step 4 — Start the Observation Journal (5 min)
Look at all three cups right now — Day 1. Nothing has happened yet! Have students draw what they see for Day 1 on their observation chart. Model saying: *"I observe that the seeds are just sitting there. The paper towel looks wet. The soil looks dry in Cup 2."* Good scientists describe what they see, even when nothing is happening yet.

Step 5 — Daily Check-Ins (Ongoing — Lessons 3–6)
Each morning for the next week, check the cups together! Use the observation chart to record what they see:

- Has anything changed? Any green poking through?
- Can you see a root growing through the clear cup?
- Draw what you observe — even if it's "nothing happened today"!

Keep the paper towel in Cup 3 **moist** (not soaking) throughout the week.

💬 Discussion Questions

🗣️ **Before planting — make a prediction:**

- "Which cup do you think will grow a plant first? Why?"

  ✨ *Both Cup 1 (Soil + Water) AND Cup 3 (No Soil + Water) should sprout! Seeds need moisture and warmth — soil helps roots anchor and feed the plant later, but isn't strictly needed to germinate. Cup 2 (Soil Only, no water) won't sprout because there's no moisture to wake the seed up.*

🗣️ **During check-ins — observe and wonder:**

- "What do you notice is different about the cups today?"

  ✨ *Guide kids to describe changes: color, size, anything poking out, roots showing through the clear cup, etc.*
- "Why do you think Cup 2 isn't growing?"

  ✨ *No water = no germination. The seed is dormant (still sleeping) because it can't activate without moisture.*
- "Cup 3 has no soil — but it's sprouting! How is that possible?"

  ✨ *Seeds don't need soil to start growing — they have their own food stored inside! Soil becomes important later for the plant to grow tall and healthy. Right now the seed just needs water, warmth, and air.*
- "What do you think will happen to the sprout in Cup 3 if it keeps growing with no soil?"

  ✨ *It will eventually run out of nutrients. Soil is what feeds a plant long-term. This is a great preview for future lessons!*

🚀 Extensions

- 🌡️ **Temperature Test:** Set up a 4th cup identical to Cup 1, but put it in a cold dark spot (like a closet or fridge). Does it grow? Explore how warmth affects germination.
- 🔦 **Light vs. Dark:** Try two identical "Soil + Water" cups — one on the sunny sill, one in a dark cabinet. Which sprouts first? (Spoiler: darkness doesn't stop germination, but affects what happens after!)
- 📚 **Read Aloud:** *"The Tiny Seed"* by Eric Carle, or *"From Seed to Plant"* by Gail Gibbons
- 🎨 **Seed Art:** Press dried bean seeds onto paper and trace their shapes. Compare shapes of different seeds (sunflower, bean, pea, corn).
- 📓 **Science Journal:** Make a simple folded-paper journal to draw the experiment results each day — good practice for scientific record-keeping!

👩‍🏫 Parent/Teacher Notes

⚠️ **Don't throw away the cups!** This experiment continues across Lessons 3–6. The cups are your ongoing data! Check them together each morning and keep updating the observation chart.

- **Label clearly:** Use a marker to write directly on the cups: *"Soil + Water"*, *"Soil Only"*, and *"No Soil + Water"* so everyone always knows which is which.
- **Use clear cups:** Being able to see the roots push through the side of the cup is one of the most magical moments for young learners. If you only have opaque cups, try pressing a seed against a wet paper towel in a zip-lock bag taped to the window instead.
- **Make it a morning routine:** Check the cups first thing each morning before other activities. Consistency builds the habit of scientific observation.
- **Refresh Cup 3:** Re-dampen the paper towel every day or two so it doesn't dry out. Cup 2 (Soil Only) stays dry on purpose — don't accidentally water it!
- **The science:** Germination requires moisture, warmth, and air. Soil provides nutrients and structure for long-term plant growth — but the seed itself contains enough stored food energy to sprout without it. That's why Cup 3 (paper towel) will sprout just like Cup 1! Cup 2 will stay dormant the whole week.
- **Timing:** Most bean seeds germinate in 3–7 days under warm, moist conditions. If it's taking longer, check that the seeds aren't waterlogged (too much water = no air) and that the space is warm enough.

🌱 Planting Day Worksheet 🌱

Name: _____ Date: _____

🖍️ Draw the Three Cups

Draw what each cup looks like on Planting Day!

Cup 1
Soil + Water 💧

draw here

Will it grow? **YES** **NO**

Cup 2
Soil Only 🏜️

draw here

Will it grow? **YES** **NO**

Cup 3
No Soil + Water 🌊

draw here

Will it grow? **YES** **NO**

*Circle **YES** or **NO** for each cup — these are your predictions! (Make your best guess!)*

📋 My Observation Chart

Each day, draw what you see in each cup. Even if nothing changed — draw that too! Good scientists always record their observations.

Day	☁️ Cup 1: Soil + Water	🏜️ Cup 2: Soil Only	💧 Cup 3: No Soil + Water
Day 1 Today!	*draw what you see*	*draw what you see*	*draw what you see*
Day 2	*draw what you see*	*draw what you see*	*draw what you see*
Day 3	*draw what you see*	*draw what you see*	*draw what you see*
Day 4	*draw what you see*	*draw what you see*	*draw what you see*
Day 5	*draw what you see*	*draw what you see*	*draw what you see*
Day 6	*draw what you see*	*draw what you see*	*draw what you see*
Day 7	*draw what you see*	*draw what you see*	*draw what you see*

💭 What did I learn?

After a week, complete these sentences:

The cup that grew first was: _____

The cup that did NOT grow was: _____

I think seeds need: _____

🔑 PARENT ANSWER KEY — Detach before giving worksheet to students

🌱 Prediction Answers

- **Cup 1 (Soil + Water) → YES, it will grow.** This is the control — soil, water, warmth, and air. All conditions for germination are met.
- **Cup 2 (Soil Only, no water) → NO, it will NOT grow.** No moisture = the seed stays dormant. It might look fine, but nothing will happen.
- **Cup 3 (No Soil + Water) → YES, it will grow!** This is the surprising one. Seeds contain their own stored food energy — they don't need soil to germinate! The wet paper towel provides moisture, the windowsill provides warmth, and air does the rest.

📋 What to Expect in the Observation Chart

- **Days 1–2:** Nothing visible yet. Seeds are absorbing water and beginning to swell. Cup 2 seeds look unchanged.
- **Days 3–5:** Cups 1 and 3 should begin showing root tips (white, thread-like). Cup 3 roots may be visible through the clear cup wall against the paper towel. Cup 2: still nothing.
- **Days 5–7:** Shoots (green stems) begin emerging in Cups 1 and 3. Roots are increasingly visible in Cup 3 and may show through the clear cup walls in Cup 1. Cup 2 remains dormant.

✅ What Success Looks Like

- Student can predict (with reasoning) which cups will grow.
- Student records observations each day — even "no change" is a valid scientific observation.
- Student can explain: *"Seeds need water and warmth to grow. Cup 2 didn't grow because it had no water."*
- Student is surprised (and delighted!) that Cup 3 sprouted without any soil — this is the "aha" moment of the lesson!

🔬 Sentence Completions (Sample Answers)

- **"The cup that grew first was…"** → Cup 1 or Cup 3 (both should sprout; Cup 3 sometimes sprouts faster because roots can be seen immediately)
- **"The cup that did NOT grow was…"** → Cup 2 (Soil Only, no water)
- **"I think seeds need…"** → water (and warmth / air)

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🌱 Parts of a Plant: Root to Flower 🌸

Life Science — Unit 2, Lesson 3 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🌿 **Subject:** Life Science — Plants & Seeds 🧰 **Kit Lesson:** No

Every plant has four main parts — and each part has its own very important job!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will investigate parts of a Plant: Root to Flower, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Plants aren't just pretty — they're amazing living machines! Every part of a plant has a **job** that helps the plant grow, drink water, make food, and create new seeds. In this lesson, your child will dig up a real plant, observe all four parts up close, and discover what each part does. Then they'll set up a colorful experiment to actually *watch* the stem carry water!

🔬 The Science

Plants have four main parts: **roots**, a **stem**, **leaves**, and a **flower**. Each part has a specific function that helps the plant survive:

- 🌾 **Roots** anchor the plant in soil and absorb water and nutrients.
- 🪴 **Stem** acts like a pipe or straw — it carries water and nutrients from the roots up to the rest of the plant.
- 🍃 **Leaves** catch sunlight and use it (along with air and water) to make food for the plant — a process called photosynthesis.
- 🌸 **Flower** attracts bees and butterflies that help the plant make seeds so new plants can grow.

Today's experiments let kids *see* two of these jobs in action: they'll observe the roots of a real plant, and watch a celery stem actually transporting colored water right before their eyes!

💡 **Key vocabulary:** roots, stem, leaves, flower, nutrients, absorb, transport, photosynthesis

📦 Materials

🌿 For the Plant Dig Activity

- 1 small weed or plant from the yard (dandelion is *perfect* — deep taproot, all 4 parts visible)
- A spoon or small garden trowel for digging
- Sheet of white paper (to lay the plant on and observe)
- Crayons or colored pencils (to sketch the plant)

🥬 For the Celery Stem Experiment

- 1 celery stalk (with leaves still on, if possible)
- 1 clear cup or glass
- Water
- Red or blue food coloring (10–15 drops — more is better!)

🔬 Let's Get Started

Step 1 — Set Up the Celery Experiment FIRST! (3 min)
Before anything else, get your celery experiment going — it needs time to work while you do the rest of the lesson!

- Fill the cup about halfway with water
- Add 10–15 drops of food coloring and stir — make it nice and bright
- Place the celery stalk in the colored water (cut the bottom of the stalk at an angle first for better absorption)
- Set it in a sunny spot and say: *"Let's check on this later and see what happens!"*

Step 2 — Warm-Up: What Do Plants Need? (5 min)
While you head outside, talk about what the kids already know:

- *"What do YOU need to survive? Water, food, air, sunlight…"*
- *"Do plants need the same things? How do you think they get them?"*
- *"Does a plant have a mouth? A stomach? How does it drink water without those things?"*

Let kids share their ideas — there are no wrong answers yet!

Step 3 — Dig It Up! Plant Excavation (10–12 min)
Head outside and find a small weed or dandelion. Together:

- Gently dig around the base, going several inches down to get the full root
- Carefully pull the whole plant out and shake off excess soil
- Lay it on the white paper and look at it together
- Ask: *"Can you find all four parts? Where are the roots? The stem? The leaves? Any flowers?"*
- Touch each part — roots feel different from stem, leaves feel different from roots
- Optional: sketch the plant with crayons, labeling each part

💡 **Dandelion tip:** Dandelions are ideal because their long taproot makes the root system super obvious. You can see all four parts clearly, and the fluffy seed head is a bonus conversation starter about how seeds spread!

Step 4 — Introduce the Four Parts (5–8 min)
Back inside (or outside on a blanket), go through each plant part:

- 🌾 **Roots:** Point to the roots. *"These are like the plant's straws AND its anchor. What do you think they do?"* (See Discussion Question below)
- 🪴 **Stem:** Run a finger up the stem. *"This is the highway — water travels all the way from the roots to the leaves through here!"*
- 🍃 **Leaves:** Hold a leaf up to the light. *"Leaves catch sunlight like little solar panels to make the plant's food."*
- 🌸 **Flower:** *"Flowers make seeds so new plants can grow — and bees love them!"*

Use the uprooted plant as a visual the whole time.

Step 5 — Check the Celery! (5 min)
Return to your celery experiment. After 30–60 minutes, you should start to see color in the leaves. (After a few hours it'll be very vivid — check again at lunch or dinner!)

- Look at the leaves — any color yet? What about the stalk?
- Optional: carefully cut the celery stalk crosswise and look at the tiny colored dots — those are the tubes that carry water!
- *"See how the colored water traveled UP the stem? That's exactly what a real stem does every single day!"*

Step 6 — Worksheet & Wrap-Up (5–8 min)
Complete the Student Worksheet (page 2). Kids label the four plant parts on the diagram and complete the job-matching activity. Finish with the discussion question below!

💬 Discussion Question

🗣️ **Ask this together:**

- **"What job do you think roots do for the plant?"**

  > ➤ *Roots are like straws — they drink up water and nutrients from the soil and send them up through the stem to the rest of the plant! They're also like anchors: they hold the plant in the ground so it doesn't blow away in the wind or wash away in the rain.*

- **"Why do you think leaves are green and flat? What's the advantage of that shape?"**

  > ➤ *Flat and wide = more surface area to catch sunlight! Green comes from chlorophyll, the special material inside leaves that actually captures the sun's energy to make food.*

- **"If a plant had no roots, what would happen?"**

  > ➤ *It couldn't drink water, so it would dry out and die. It also couldn't stay in the ground — the wind would knock it over! Roots do two important jobs at once.*

🚀 Extensions

- 🌹 **White Carnation Upgrade:** Try the same food-coloring experiment with a white carnation — the result is even more dramatic and beautiful!
- ✂️ **Split-Color Celery:** Carefully split the celery stalk partway up and put each half in a different color of water. What happens?
- 🗺️ **Neighborhood Plant Hunt:** Walk around the block and find 5 different plants. Do they all have the same four parts? Can you find roots, stem, leaves, and flowers on each?
- 🎨 **Nature Journal:** Press a small leaf between two sheets of paper and rub over it with a crayon (leaf rubbing) — try several different leaf shapes and compare!
- 📚 **Read Aloud:** *"From Seed to Plant"* by Gail Gibbons, or *"The Tiny Seed"* by Eric Carle
- 🫘 **Bean Check:** How are the bean plants from Lesson 2 doing? Any sprouts? Can you identify the stem and leaves on the seedling?

👩‍🏫 Parent/Teacher Notes

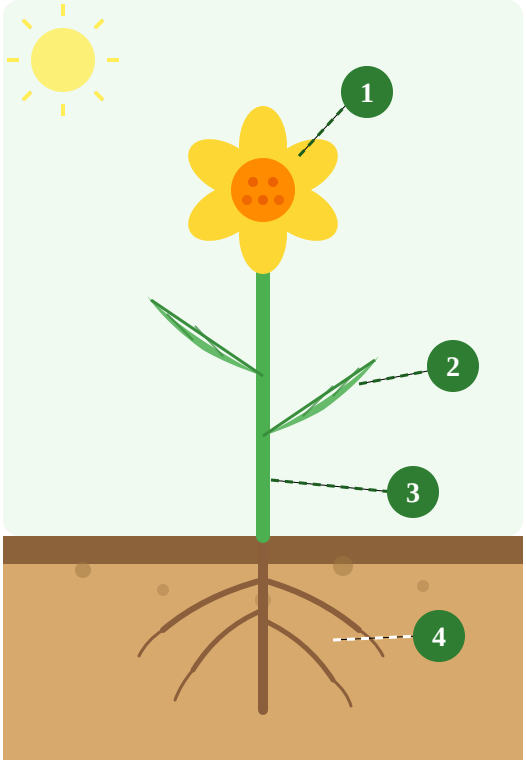
- ⏱️ **Start the celery FIRST:** The food-coloring experiment needs time to work — set it up at the very beginning so the color has 30–60 minutes to travel up while you do the outdoor plant dig. Check it mid-lesson and again after. Overnight is even more impressive!
- 🌹 **Carnations are even better:** If you can grab white carnations from a grocery store, use those instead of celery. The color change is vivid, dramatic, and kids go absolutely wild for it. Still works great with celery if that's what you have.
- 🌼 **Dandelions are the ideal dig plant:** Their deep taproot makes the root system super obvious — kids can actually see the whole root structure, and it's hard to miss any of the four parts. Dandelions are everywhere and free!
- 🫘 **Check in on Lesson 2 bean plants:** While you're doing this lesson, peek at any beans you planted in Lesson 2. Any sprouts? If a seedling has emerged, use it as a live example — this is a real plant with real parts right in front of them!
- 📖 **Vocabulary to reinforce:** *roots*, *stem*, *leaves*, *flower*, *nutrients*. Use these words naturally throughout the lesson, not just as definitions.
- 🔬 **The celery cross-section is optional but cool:** If your kids are curious, cut the celery crosswise with a sharp knife (adult only) and look at the tiny colored dots in the stalk. Those are the vascular tubes that transport water — a great visual for "pipes inside the stem!"
- ✅ **Assessment:** Can the child name all four plant parts? Can they point to each one on a real plant or diagram? Can they tell you one job for each part? That's success for Kindergarten.

🌱 Parts of a Plant Worksheet 🌸

Name: _____ Date: _____

🏷️ Part 1: Label the Plant!

Look at the numbers on the plant diagram. Write the correct plant part name on each blank line. Use the Word Bank to help you!



Write the name for each numbered part:

1 _____

2 _____

3 _____

4 _____

✏️ **Word Bank**

roots · stem · leaves · flower

🔗 Part 2: Match the Job!

Draw a line from each plant part on the LEFT to its job on the RIGHT!

🌾 **ROOTS**

🪴 **STEM**

☀️ Catches sunlight to make food for the plant

🌸 Makes seeds and attracts bees & butterflies

🍃 **LEAVES**

🌸 **FLOWER**

💧 Drinks up water from soil & holds the plant in the ground

🚿 Carries water from the roots all the way up to the leaves

🎨 **Bonus:** Color the plant diagram above! Use green for the leaves and stem, yellow for the flower, and brown for the roots.

✂️ PARENT / TEACHER ANSWER KEY — Remove before giving worksheet to student

🏷️ Part 1: Label the Plant — Answers

① **Flower** — the yellow bloom at the top

② **Leaves** — the flat green parts on the sides

③ **Stem** — the vertical green stalk in the middle

④ **Roots** — the brown branching parts underground in the soil

🔗 Part 2: Match the Job — Answers

🌾 **ROOTS** → 💧 Drinks up water from soil & holds the plant in the ground

🪴 **STEM** → 🚿 Carries water from roots up to the leaves

🍃 **LEAVES** → ☀️ Catches sunlight to make food for the plant

🌸 **FLOWER** → 🌸 Makes seeds and attracts bees & butterflies

💡 Deeper Notes for Parents

- **Flower → seeds:** Some kids initially think flowers are "just pretty" — reinforce that the flower's real job is reproduction. Bees carry pollen from flower to flower, which lets the plant make seeds. That's how new plants grow!
- **Roots → two jobs:** Roots drink AND anchor. Both are correct answers. If a child says both, that's excellent scientific thinking — reinforce it.
- **Leaves → photosynthesis:** At Kindergarten level, "catches sunlight to make food" is the right level of detail. If they ask how, you can say: "Leaves have a special ingredient called chlorophyll that turns sunlight, air, and water into sugar — the plant's food!"
- **Stem → vascular tissue:** The celery experiment visually demonstrates this! The colored dots in a cross-cut celery stalk are the tiny tubes (xylem) that carry water. You don't need the vocabulary — just point and say "those are the water pipes!"

✅ What Success Looks Like

- Child can name all four plant parts when shown a real plant or diagram.
- Child can describe at least one job for each part in their own words.
- Child connects the celery experiment to what stems actually do.
- Bonus: Child notices that both roots and flowers serve the plant's survival (water + reproduction).

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🌞 Sunlight or No Sunlight? 🌱

Life Science — Unit 2, Lesson 4 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🌿 **Subject:** Life Science — Plants & Seeds 🧰 **Kit Lesson:** No

Two plants. Same water. But only one gets sunlight — what do you think will happen?

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will observe what happens when a plant doesn't get sunlight and conclude that plants need light to survive.

🎯 The Big Idea

Plants are amazing — they make their own food! But to do that, they need **sunlight**. In this lesson, your kids will set up a simple experiment: one plant in a sunny spot, one in the dark. Over 3–5 days, they'll observe what happens and discover firsthand why sunlight is so important for plants.

🔬 The Science

Plants use sunlight, water, and air to make their own food — this process is called **photosynthesis**. It happens inside the green parts of a plant (the leaves!). When a plant doesn't get enough sunlight, it can't make food. It gets weak, pale, and stretched out — reaching desperately toward any light it can find.

A plant left in complete darkness will start to look very different from one in a sunny window: it turns pale yellow or white, grows tall and spindly (scientists call this *etiolated*), and eventually droops. It's not dead — but it's definitely not healthy.

💡 **Key vocabulary:** photosynthesis · sunlight · energy · survive · observe

📦 Materials

🌱 Plants

- Two small potted plants (roughly the same size) *or* use the bean sprouts from Lesson 2 if they've grown!
- If using bean sprouts: pick two cups that look similar in size and health

🏠 Setup Supplies

- A sunny windowsill or bright spot (Plant A)
- A dark closet — or even better, a cardboard box with a lid (Plant B)
- Water & a small cup for watering
- Observation journal or a blank notebook
- Pencil and crayons for drawing observations

🔬 Let's Get Started

Step 1 — Set Up the Experiment FIRST (5 min)
Before anything else, get the experiment going — it needs 3–5 days to work!

- Label two plants (or cups): **"Sunny"** and **"Dark"**
- Place the "Sunny" plant on the brightest windowsill you have
- Put the "Dark" plant in a closet, or better yet, inside a cardboard box with the lid closed
- Water *both* plants the same amount right now

Say: *"We're going to let these plants sit for a few days and then see what happened. We're scientists making a prediction — what do you think will happen to the dark plant?"*

Step 2 — Make a Prediction (5 min)
In their observation journal, kids draw both plants as they look *right now* (they should look the same!). Ask:
"Draw what you think each plant will look like in 3 days. Will they be the same? Or different?"
Write or dictate their prediction underneath.

Step 3 — Talk About What Plants Need (10 min)
While the experiment runs, discuss! Use the questions below. Cover the key ideas:

- Plants need water, air, and **sunlight** to survive
- Sunlight gives plants the energy to make their food (photosynthesis!)
- Without sunlight, a plant can't make food — and gets weak

Try this: turn off the lights in the room for a moment. Say: *"Imagine living in the dark all the time — with no food. How would you feel? That's how the plant in the box feels!"*

Step 4 — Daily Check-Ins (over 3–5 days)
Each day, take a quick peek at the sunny plant and open the box for the dark plant. Ask kids:

- *"Has anything changed?"*
- *"What color is it? Is it standing up straight or drooping?"*

Draw a quick sketch in the observation journal each day. Water both plants the same amount.

Step 5 — Big Reveal: Side-by-Side Comparison (in Lesson 5 or 6)
After 3–5 days, bring both plants out and put them next to each other. Let the kids look closely. Then complete the Student Worksheet: draw both plants side by side and circle which one looks healthier.

Step 6 — Wrap Up (5 min)
Ask: *"So what do plants need that the dark plant wasn't getting?"* Guide kids to say: **SUNLIGHT!**
Introduce the word *photosynthesis*: *"Plants make their own food using sunlight — that's called photosynthesis. Pretty cool, right?"*

💬 Discussion Questions

🗣️ **Question 1:** "The plant in the dark is still alive — but does it look healthy? What's different?"

💬 *The dark plant looks pale and stretched out because it's reaching for light. Plants need sunlight to make their food through their leaves — that's photosynthesis. Without it, they get weak and can't stay healthy. They're still alive, but just barely hanging on!*

🗣️ **Question 2:** "What would happen if all the clouds blocked the sun for a whole year?"

💬 *Without sunlight, plants couldn't make their food and would die. Without plants, animals that eat plants would lose their food. Even meat-eating animals would eventually run out of food too. Everything in nature is connected — sunlight is the very beginning of almost all food on Earth! (This is called a food chain — a fun preview for later lessons.)*

🚀 Extensions

- 🌀 **Partial Light Test:** Try a third plant — put it in a dim corner (not totally dark, not sunny). Where does it end up compared to the other two?
- 🌿 **Leaf Rubbings:** Place a leaf under paper and rub a crayon over it. Look at the veins — those carry water and sugars made by photosynthesis!
- 📚 **Read Aloud:** *"A Seed Is Sleepy"* by Dianna Hutts Aston or *"From Seed to Plant"* by Gail Gibbons
- 🎨 **Color Sorting:** Look at pictures of healthy plants (green, upright) and unhealthy plants (yellow, droopy). Sort them: "Enough sunlight" vs. "Not enough sunlight"
- 🔭 **Grow Toward the Light:** Put a plant near a window for a week and watch it lean toward the light. This is called *phototropism* — another bonus word for curious kids!

👩‍🏫 Parent/Teacher Notes

- ⚡ **Start the experiment FIRST:** Do Step 1 at the very beginning of the lesson — the experiment takes 3–5 days to show results. Final observations happen in Lesson 5 or 6, so plan accordingly.
- 📦 **Box beats closet:** A cardboard box with a lid works better than a closet — it's completely dark, stays the same temperature, and is easy for kids to peek inside without flooding the plant with light each time.
- 🫘 **Lesson 2 connection:** Check on the 3-cup experiment from Lesson 2 (seed, no water, no soil). Which cups have sprouted? Compare notes — seeds need water to germinate, but plants need sunlight to keep growing.
- 🔬 **Introduce photosynthesis gently:** Don't shy away from the big word! Say it a few times and make it fun: *"Photo means light, and synthesis means making — so photosynthesis means making food with light!"*
- 🌿 **Fun bonus word:** The pale, leggy, stretched-out look of a plant grown in the dark has a scientific name: *etiolated*. You don't need to teach it, but curious parents may enjoy knowing it!
- ✅ **Assessment:** Can the child explain that plants need sunlight to stay healthy? Can they point to the healthier plant and say why it looks better? That's the win.

🌞 My Plant Observation 🌱

Sunlight or No Sunlight? — Worksheet

Name: _____ Date: _____

After 3–5 days, draw what each plant looks like. Then answer the questions below!

☀️ Sunny Plant

This plant was on the windowsill

Draw here ✏️

🌑 Dark Plant

This plant was in the dark box

Draw here ✏️

🔍 What did you notice?

Write or tell one difference between the two plants:

_____

_____

_____

Circle the plant that looks healthier:

☀️ Sunny Plant or 🌑 Dark Plant

Why does the plant need sunlight? (write, draw, or dictate)

_____

_____

🌿 **Big word I learned today:** **Photosynthesis** — plants make their own food using sunlight! ☀️🍃

📋 PARENT / TEACHER ANSWER KEY — Detach before giving worksheet to students

🌿 What to Look For After 3–5 Days

✅ The **Sunny Plant** should look green, upright, and healthy — maybe even a little bigger

✅ The **Dark Plant** will look pale (yellow or white), stretched tall and spindly, and possibly droopy or leaning

✅ Both were watered equally — the only variable was sunlight

☀️ Why the Dark Plant Looks Different

Without sunlight, the plant cannot perform **photosynthesis** — the process by which it makes its own food using light energy. The plant stretches toward any available light source (a behavior called *phototropism*), causing it to grow tall and thin. It loses its green color (chlorophyll breaks down without sunlight) and becomes pale. The plant is alive, but weak and unable to thrive.

✅ Worksheet Expected Answers

✅ One observed difference (any valid observation): "The dark plant is yellow," "The sunny plant is greener," "The dark plant is taller and droopy," "The sunny plant looks stronger"

✅ **Circle the healthier plant:** ☀️ Sunny Plant

✅ Why does the plant need sunlight: "To make food" / "To stay green and strong" / "So it can grow"

💬 Discussion Question Answers

Q1 — "The plant in the dark is still alive — but does it look healthy?"
Expected: No! The dark plant looks pale, stretched, and droopy. Plants need sunlight to make their food (photosynthesis), so without light they get weak even if they still have water.

Q2 — "What would happen if all the clouds blocked the sun for a whole year?"
Expected: Plants would die without sunlight. Then animals that eat plants would lose their food. Eventually all living things would be affected. Sunlight is the energy source at the base of almost all food chains on Earth.

🔬 Vocabulary Check

✅ **Photosynthesis:** The process plants use to make food from sunlight, water, and air

✅ **Sunlight / Energy:** The source of power plants need to run photosynthesis

✅ **Survive:** To stay alive and healthy; what plants (and animals!) are trying to do

✅ **Etiolated (bonus!):** The scientific word for a pale, leggy plant grown without enough light

✅ What Success Looks Like

✅ Child can correctly identify the sunny plant as healthier

✅ Child can describe at least one visible difference between the two plants

✅ Child can say (in their own words) that plants need sunlight to survive

✅ Bonus: Child can use (or recognize) the word "photosynthesis"

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🌱 Seed Travelers: How Seeds Move 🌬️

Life Science — Unit 2, Lesson 5 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🌿 **Subject:** Life Science — Plants & Seeds 🧰 **Kit Lesson:** No

Wind, water, animals, and even explosions — seeds are born to travel!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will investigate seed Travelers: How Seeds Move, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Seeds don't just fall straight down and stay there — they're built to **travel**! Plants have clever tricks for sending their seeds far away: riding the wind, floating on water, hitching a ride on animals, or even shooting out like tiny cannons. The farther a seed travels, the better its chances of finding a good spot to grow.

🔬 The Science

When a seed lands too close to its parent plant, the two plants end up competing for the same water, sunlight, and space. That's a problem! Plants can't pick up and walk away, so over millions of years they developed incredible **adaptations** — special features that help seeds disperse (spread out) to new locations.

💨 **Wind** Feathery, papery, or winged seeds catch the breeze. Dandelion fluff, maple helicopters, cattail fluff.

🌊 **Water** Buoyant seeds float downstream or in ocean currents. Coconuts are champion water travelers!

🐿️ **Animals** Sticky burrs hitch rides on fur & socks. Tasty fruits get eaten and seeds come out far away!

💥 **Explosion** Some seed pods build up pressure and BURST open, flinging seeds through the air. Touch-me-nots!

💡 **Key vocabulary:** dispersal, germination, adaptation, seed pod, buoyant, burr

📦 Materials

🌿 Seeds & Nature Items

- Dandelion puffs (or cotton balls with a few small seeds glued on)
- Maple seeds / samaras ("helicopters") — or paper helicopter templates
- Small burrs or sticky seeds (burdock, foxtail, etc.) — or small Velcro pieces
- Assorted kitchen seeds: sunflower seeds, dried beans, apple seeds

🧰 Activity Supplies

- One or two fuzzy socks (for the animal dispersal walk-around)
- A bowl of water (for the float test)
- Paper strip for making helicopter spinners (approx. 1" × 6")
- Scissors, crayons, pencils
- Glue stick (for bonus seed-gluing activity on worksheet)

🔬 Let's Get Started

Step 1 — Seed Hunt (5–10 min, outdoors if possible!)
Head outside to the yard or a nearby park. Challenge kids: *"How many different seeds can you find in 5 minutes?"*
Collect them in a cup or on a piece of tape. Look for dandelion puffs, burrs on socks/pant legs, acorns, berries, seed pods.
When you come back inside, spread them on a table and look closely.
Ask: *"Did any seeds come inside without you meaning to bring them? How did they travel?"*

Step 2 — Introduce the Four Travelers (5 min)
Hold up or point to examples of each travel method. Say the name and do a motion together:

- 🌬️ **Wind** — blow on a dandelion puff or toss the maple helicopter up and watch it spin
- 🌊 **Water** — mime a seed floating downstream (arms out, drift sideways)
- 🐿️ **Animals** — stick a burr to your sleeve and say "Oops! Free ride!"
- 💥 **Explosion** — crouch tiny, then burst arms out and jump — POP!

Step 3 — Wind Station (5 min)
Place a cotton ball "seed" or dandelion puff at one end of the table. Each child takes a turn blowing it as far as they can in one breath.
Then drop a paper helicopter from shoulder height and watch it spin down. Try from a higher spot (standing on a step stool safely, or just dropping it from a higher floor if available).
"Why do you think maple seeds have those little wings?"

Step 4 — Animal Dispersal Station (5 min)
Pull a fuzzy sock over one hand or foot. Drop a few small burrs (or tiny Velcro squares) on the floor/table. Walk your "sock foot" through them. Check what stuck!
"Imagine you're a deer walking through a field. Where might those seeds end up?"
Carefully pull them off and examine them — most burrs have tiny hooks (nature's Velcro).

Step 5 — Water Float Test (5 min)
Fill a bowl with water. Drop in a variety of seeds: sunflower seeds, dried beans, apple seeds, a small piece of coconut shell if available, cotton ball.
Observe: Which seeds float? Which sink? Record on fingers: *"Floaters could travel by water. Sinkers might use a different method."*
"If a seed floated into a river and came out on a muddy bank 10 miles away — what would happen next?"

Step 6 — Explosion Demo (3 min)
Show photos or a video of touch-me-not (Impatiens) seed pods popping, or squirt a ripe raspberry gently over a piece of white paper — seeds fly out!
A water balloon popped at arm's length gives a great feel for the sudden force. Say: *"The pod builds up pressure inside, just like a balloon, until — POP! Seeds fly out in all directions!"*

Step 7 — Wrap Up & Worksheet (10 min)
Come together and list the four dispersal methods from memory. Then move to the Student Worksheet — match seeds to their travel method, draw three types of seed travel, and (bonus) glue real seeds next to the correct label.

💬 Discussion Questions

🗣️ **Question 1:** "Why is it important for seeds to travel *away* from the parent plant?"

💡 *Suggested answer: If all the seeds fell right next to the parent plant, they'd be too crowded — fighting for water, sunlight, and space. Spreading out gives each seed a better chance to grow into a healthy plant.*

🗣️ **Question 2:** "What do you think would happen if all the seeds from a big oak tree fell in exactly the same spot and none of them moved?"

💡 *Suggested answer: The seeds would all sprout in the same tiny area. The seedlings would grow so crowded that they'd compete for water, sunlight, and nutrients — and most of them would die. Only one or two might survive. Spreading out means more plants get to grow up healthy!*

🗣️ **Bonus question:** "Can you think of any seeds that travel in more than one way?"

💡 *Suggested answer: Yes! Coconuts can float in the ocean (water) AND get moved by animals or people. Berries travel by animal (eating) but could also float in a stream. Some seeds are so light they can be blown by wind AND float on water. Plants often have "backup plans" for dispersal!*

🚀 Extensions

- 🔍 **Microscope Burr:** Look at a burr under a magnifying glass. Can you see the tiny hooks? Compare it to a piece of Velcro under the same magnification.
- 📏 **Wind Distance Test:** Mark a start line. Each child blows a cotton ball seed three times and measures how far it goes. Whose seed traveled farthest? What helped?
- 🎨 **Seed Watercolor:** Paint a seed journey — draw the seed at home on the parent plant, mid-travel (flying, floating, riding), and at its new home sprouting.
- 📚 **Read Aloud:** *"A Seed Is Sleepy"* by Dianna Hutts Aston, or *"How a Seed Grows"* by Helene J. Jordan
- 🌱 **Connect to Prior Lessons:** Look back at your 3-cup water experiment and sunny/dark plant experiment — now that seeds have traveled to a new spot, what do they need to survive and grow?

👩‍🏫 Parent/Teacher Notes

- **Go outside first:** This is a wonderful outdoor lesson — do the seed hunt in the yard or park first, then come inside for the stations. Kids who find their own seeds are far more engaged when you explain how those seeds travel.
- **Best burrs for the sock activity:** Burdock burrs (the classic "nature's Velcro") are ideal. Foxtail grass and cocklebur also work well. Pull them off carefully after — they can mat in hair. Alternatively, use small pieces of hook-side Velcro as a safe classroom substitute.
- **Paper helicopter template:** Cut a strip of paper about 1" wide and 6" long. Make a small cut about 2" down the middle from the top. Fold the two resulting flaps in opposite directions — one forward, one back. Drop it from height and it will spin! Add a small paper clip to the bottom for more realistic spinning. Compare to a real maple samara if available.
- **Explosion dispersal demo:** Squirting cucumbers, touch-me-nots (Impatiens), and witch hazel are great real-world examples. In the classroom, gently squeezing a ripe raspberry over paper dramatically shows seeds flying out. A popped water balloon also illustrates the explosive "pop" concept safely.
- **Vocabulary focus for this lesson:** *dispersal* (spreading seeds to new places), *germination* (when a seed starts to grow), *adaptation* (a special feature that helps a living thing survive). Use these words naturally throughout the lesson, not just at the end.
- **Connect to previous lessons:** Reference the 3-cup water experiment and the sunny/dark plant experiment from earlier in Unit 2 — once a seed travels to a new spot, it still needs water, light, and space (just like the experiments showed) to germinate and grow.
- **Assessment:** Can each child name at least three ways seeds travel? Can they match a seed type to its dispersal method? Can they explain in their own words why spreading out is good for seeds?

🌱 Seed Travelers — Student Worksheet

Name: _____ Date: _____

🔗 Part 1: Match the Seed to Its Travel Method

Draw a line from each seed to the way it travels. (Hint: some seeds might travel more than one way — pick the best match!)

🌿 Seed / Plant	How does it travel?
🌼 Dandelion puff	_____
🍁 Maple seed (helicopter)	_____
🌰 Coconut	_____
🪢 Sticky burr	_____
🍓 Strawberry / berry	_____
💥 Touch-me-not seed pod	_____

💨 **Wind** 🌊 **Water** 🐿️ **Animals** 💥 **Explosion**

✏️ Part 2: Draw It!

Draw a seed traveling in each way below:

💨 Flying in the Wind	🌊 Floating on Water	🐿️ Hitching a Ride
Draw a seed being carried by the wind!	*Draw a seed floating on a river or ocean!*	*Draw a seed stuck to an animal (or your sock!)!*

⭐ BONUS: Glue Real Seeds!

Find real seeds and glue one in the correct box below. Which dispersal method does each seed use?

💨	🌊	🐿️	💥
Wind seed	**Water seed**	**Animal seed**	**Explosion seed**
glue here	glue here	glue here	glue here

🤔 Think & Write

What would happen if ALL the seeds from one plant fell in the exact same spot?

🔑 PARENT / TEACHER ANSWER KEY — Remove before giving worksheet to students

Part 1: Matching Answers

- 🌼 **Dandelion puff** → 💨 **Wind** (feathery parachute catches air currents)
- 🍁 **Maple seed (helicopter)** → 💨 **Wind** (samara wing spins in the breeze, slows descent, drifts far)
- 🌰 **Coconut** → 🌊 **Water** (very buoyant husk; can float in ocean for months)
- 🪢 **Sticky burr** → 🐿️ **Animals** (tiny hooks latch onto fur, feathers, and clothing)
- 🍓 **Strawberry / berry** → 🐿️ **Animals** (fruit is eaten; seeds pass through digestive system unharmed and deposited elsewhere)
- 💥 **Touch-me-not seed pod** → 💥 **Explosion** (pod builds pressure; slightest touch triggers explosive seed ejection up to 6 feet away)

💡 Notes on Tricky Answers

- **Maple seeds:** Primarily wind, but can also land in streams and float a short distance (bonus water travel!). Wind is the primary mechanism.
- **Coconut:** Primarily water — coconuts are famous ocean travelers. However, animals (including humans) also move coconuts.
- **Berries:** Animals eat them and deposit seeds far away. Some berries also fall in streams (water). Animal is primary.
- **If a child picks "two methods" for any seed, that's excellent scientific thinking!** Praise it and discuss that seeds often have more than one dispersal strategy as a backup.

✅ What Success Looks Like

- Child can name at least **three dispersal methods** (wind, water, animal, explosion).
- Child can correctly match at least **four of six seeds** to their travel method.
- Drawing activity shows understanding of each dispersal concept (seed with wings blowing, seed on water, seed on animal fur).
- Think & Write response shows awareness that crowding is bad for plant survival. Look for ideas like: "they'd fight for water," "too crowded," "not enough sun," or "only one would grow."

🌱 Think & Write — Sample Strong Answer

"If all the seeds fell in the same spot, there would be too many plants in one place. They would fight for water and sunlight and most would die. Seeds need to spread out so each one has enough room and food to grow."

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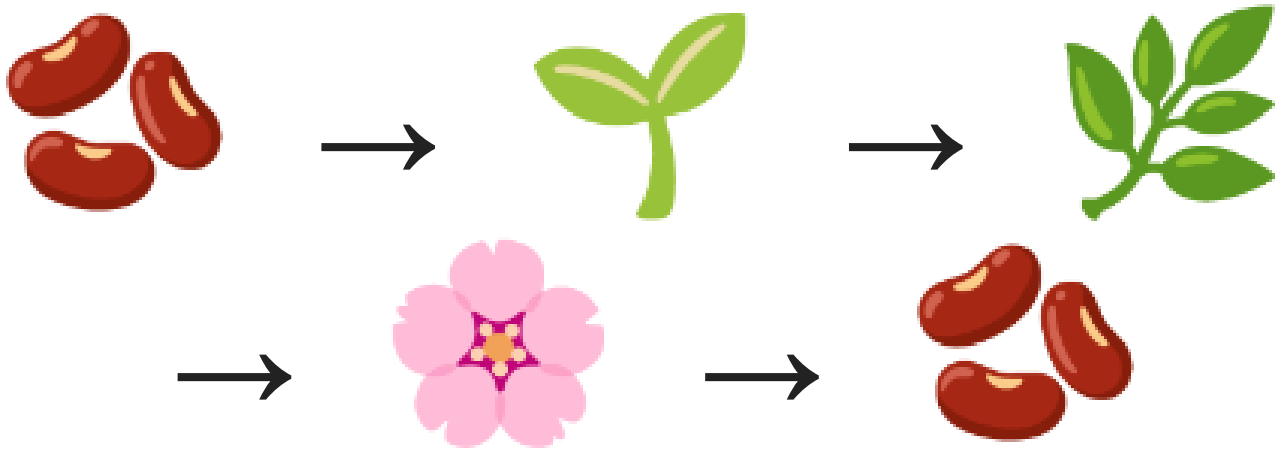
🌱 My Plant's Life Story 🌱

Life Science — Plants & Seeds — Unit 2, Lesson 6 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Life Science — Plants & Seeds 🧰 **Kit Lesson:** No



A tiny seed becomes a plant — and that plant makes new seeds. The cycle never ends!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will investigate my Plant's Life Story, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎓 Unit 2 Wrap-Up — Super Seeds & Perfect Plants

Congratulations — you finished all 6 lessons! Here's a look back at everything you explored together:

1. **What's Inside a Seed?** — Dissected beans to find the embryo, food supply, and seed coat.
2. **Let's Plant!** — Planted bean seeds in cups; learned what seeds need to germinate (water, warmth, soil).
3. **Roots & Shoots** — Observed which direction roots and shoots grow; learned roots anchor and absorb water.
4. **What Do Plants Need?** — Tested sunlight and water; tied back to K-LS1-1 (plants need sunlight, water, air, nutrients).
5. **Parts of a Plant** — Named roots, stem, leaves, flower, fruit; learned each part's job.
6. **My Plant's Life Story** — Sequenced the full plant life cycle; made a Life Cycle Wheel. *(Today!)*

📖 **Culminating Project: Our Little Garden Book**
Make a small 6-page booklet — one page per lesson! Each page can include a drawing, one or two facts, and a label. Staple together for a keepsake that shows everything learned this unit. Great for showing grandparents or friends what a scientist can do!

🎯 The Big Idea

A plant doesn't just grow and stop — it goes through a **life cycle**! A seed grows into a sprout, then into a full plant, which makes flowers, then fruit, and finally new seeds. Those new seeds grow into new plants, and the whole cycle begins again. That's why we call it a *cycle* — it goes around and around forever!

🔬 The Science

Every living thing has a **life cycle** — a series of stages it passes through from birth to making new life. For a flowering plant like a bean, the cycle has four main stages:

1. **Seed** — the beginning; contains everything the plant needs to start growing
2. **Sprout / Seedling** — the seed *germinates*; roots push down, a shoot pushes up
3. **Growing Plant** — leaves grow and make food from sunlight; the stem grows taller
4. **Flower & Seeds** — flowers attract pollinators; *pollination* leads to fruit, which contains new seeds

When the fruit ripens and falls (or is eaten!), those new seeds land in soil — and the cycle starts all over again.

💡 **Key vocabulary:** life cycle, seed, germination, seedling, sprout, growing plant, flower, pollination, fruit

📦 Materials

🌿 From Your Home / Garden

- Bean plants from Lesson 2 (about 1 week old — for observation)
- Access to a garden or yard (optional — to see plants at different stages)
- Pictures of plant life stages (optional — from a magazine or printed)

🎨 For the Life Cycle Wheel Craft

- Paper plate (1 per student)
- Crayons or markers
- Scissors ✂️
- Brass fastener or tape (optional — to make a spinning wheel pointer)

📄 For the Worksheet

- Printed worksheet (last pages of this lesson)
- Pencil, crayons, or markers
- Glue stick (if using cut-and-paste version)

🔬 Let's Get Started

Step 1 — Observe Your Bean Plant! (5 min)
Bring out the bean plants from Lesson 2. Let each child look closely at their plant. Ask:
"What do you notice? What does it look like now compared to when we planted it?"
Point out any visible roots, the stem, and the first leaves (called **cotyledons**). Ask: *"Which stage of the life cycle is your plant at right now?"*

Step 2 — Tell the Life Cycle Story (5–7 min)
Walk through the four stages together using the emoji sequence on the board or a printed poster:

- 🫘 **Seed** — "Remember when we dissected the seed? It had a tiny baby plant inside!"
- 🌱 **Sprout** — "The seed drank water and warmth — then *germination* happened! The baby plant woke up."
- 🌿 **Growing Plant** — "Leaves grew to catch sunlight. Roots drank water. The plant got bigger!"
- 🌸 **Flower & Seeds** — "Flowers attract bees and butterflies for *pollination*. That makes fruit! Inside the fruit — new seeds."

Then say: *"And guess what happens next? Those seeds fall to the ground and… the whole story starts AGAIN!"*

💡 **Pro tip:** Do a slow dramatic pause before "the whole story starts AGAIN!" — kids love the reveal. You can even spin your hand in a big circle to show "cycle."

Step 3 — Make the Life Cycle Wheel (10–15 min)
Give each child a paper plate. Help them fold it into four equal sections (like a pizza).

- Open it back up — now you have 4 sections marked
- Label each section (or let them copy): **1 Seed, 2 Sprout, 3 Growing Plant, 4 Flower & Seeds**
- Draw or color each stage in its section — encourage them to look at their real bean plant for reference!
- Draw arrows along the edge going clockwise (to show the cycle keeps going)
- **Optional:** Add a brass fastener in the center with a small cardstock pointer for a spinning wheel

"Use your pointer to show me — where is YOUR plant right now?"

Step 4 — Garden Walk (optional, 5–10 min)
If weather permits, take a walk outside! Look for plants at different stages — maybe a seed pod on the ground, a sprout coming up, a plant in full bloom, or a fruit forming on a vine.
"Can you find a plant in each stage of the life cycle?"

Step 5 — Complete the Worksheet (10 min)
Use the circular life cycle diagram worksheet. Students draw each stage in its section, add arrows, and complete the sentence at the bottom. Younger students can dictate their sentence.

Step 6 — Big Discussion & Wrap-Up (5 min)
Come together for the discussion questions below. Celebrate finishing all 6 lessons of Unit 2!
"You are now officially a Plant Scientist. 🌿 What was your favorite experiment?"

💬 Discussion Questions

🗣 **"Why is it called a *cycle* and not just a *line*?"**

Because it goes around and around! When a plant makes seeds, those seeds grow into NEW plants that make MORE seeds. It never really ends — it circles back to the beginning.

🗣 **"What would happen if a plant never made seeds?"**

There would be no new plants! The cycle would stop. Seeds are how plants make more plants and keep going.

🗣 **"What does a seed need to start its life cycle? Can you remember from Lesson 2?"**

Water, warmth, and soil — those help germination happen! Then once leaves grow, the plant also needs sunlight and air.

🗣 **"Do YOU have a life cycle? Do all living things?"**

Yes! Humans are born, grow up, and have babies — that's a life cycle too! All living things have one — dogs, trees, butterflies, even mushrooms.

🚀 Extensions

- ⏱️ **Fast vs. Slow:** A bean completes its life cycle in about 60 days. An oak tree takes *decades* just to grow big enough to make acorns. Compare them on a timeline!
- 🦋 **Other Life Cycles:** Look up the butterfly life cycle — egg → caterpillar → chrysalis → butterfly. How is it similar to a plant's? How is it different?
- 📚 **Read Aloud:** *"The Tiny Seed"* by Eric Carle, *"Jack's Garden"* by Henry Cole, or *"From Seed to Plant"* by Gail Gibbons
- 🌱 **Transplant Ceremony:** Move your bean plants to a bigger pot or outside in the garden. Mark the date — and predict when it will flower!
- 📖 **Garden Book:** Start the Our Little Garden Book project — one page per lesson, a keepsake of the whole unit.

👩‍🏫 Parent/Teacher Notes

- **Culminating lesson:** This is Lesson 6 of 6 — take a moment to look back together at all 6 experiments! Ask: *"What did we learn about seeds? About roots? About what plants need?"* This review ties directly back to K-LS1-1 (plants need sunlight, water, air, and soil/nutrients).
- **Transplant ceremony:** The bean plants from Lesson 2 should be ready to move to a bigger pot or outside. Make it special! Let each child carefully dig a hole and plant their bean with a little ceremony. Science + ritual = memory.
- **Life cycle as a concept:** Introduce the phrase "life cycle" clearly — all living things have one. You might point to a butterfly book, a caterpillar, even your pet or a tree in the yard. The concept is huge and will come up in every science unit from here on.
- **Vocabulary to introduce today:**

  Life cycle — the series of stages a living thing passes through

  Seedling — a very young plant that just sprouted

  Fruit — the part of the plant that holds seeds (tomatoes, apples, bean pods!)

  Germination — when a seed starts to grow (wakes up!)

  Pollination — when pollen moves from flower to flower (bees help!)

- **Extension — fast vs. slow life cycles:** A bean plant goes from seed to seed in about 60 days. A sunflower takes about 80 days. An oak tree can take 20–50 years to produce its first acorn. This is a great way to introduce the concept of variation in life cycles.
- **Garden Book project:** Encourage families to make a small 6-page booklet — one page per lesson — as a culminating project. Each page: a drawing, one sentence, and a label. Stapled together, it becomes a keepsake science journal. Perfect for grandparents!
- **K-LS1-1 wrap-up:** Circle back to the standard — *what do plants need to survive?* By now, students have observed it in multiple experiments: sunlight (Lesson 4), water (Lesson 2 & 4), air (Lesson 4), soil/nutrients (Lesson 2 & 3). Ask them to list it from memory — they'll likely get all four.

🌿 Plant Life Cycle Wheel 🌿

Name: _____ Date: _____

Draw each stage of the plant life cycle in its box. Add arrows to show the cycle keeps going!

2

🌱

Sprout

Draw a sprout coming out of the ground →

1

🫘

Seed

Draw a seed in the ground

🔄 Life Cycle

3

🌿

Growing Plant

Draw a plant with leaves

↓

4

🌸

Flower & Seeds

Draw a flower with fruit/seeds

Seed → Sprout → Growing Plant → Flower & Seeds → back to Seed!

✏️ **Finish this sentence:**

The plant life cycle is called a *cycle* because _____

_____

🌱 My Plant Today

Look at your bean plant from Lesson 2. Which stage is it at? Circle it:

🫘 Seed 🌱 Sprout 🌿 Growing Plant 🌸 Flower & Seeds

✏️ Draw your real plant here and write (or tell) one thing you notice:

🌿 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

✅ Correct Life Cycle Order

The four stages should appear in this clockwise order on the wheel:

1. 🫘 **Seed** — A seed contains a tiny embryo plant and stored food, protected by a seed coat.
2. 🌱 **Sprout (Seedling)** — With water and warmth, the seed germinates: roots push down, a shoot pushes up, and the first leaves emerge (cotyledons).
3. 🌿 **Growing Plant** — The plant grows true leaves, which capture sunlight for photosynthesis. The stem grows taller; roots grow deeper.
4. 🌸 **Flower & Seeds** — Flowers attract pollinators (bees, butterflies). After pollination, fruit develops. Inside the fruit are new seeds — and the cycle begins again!

Key point: Arrows should go *clockwise* from Seed → Sprout → Growing Plant → Flower & Seeds → back to Seed.

✅ Sentence Completion — Accepted Answers

Sample correct responses for: *"The plant life cycle is called a cycle because ___"*

- "…it goes around and around and never stops."
- "…the seeds make new plants that make more seeds."
- "…it keeps going in a circle, back to the beginning."
- "…it never has an ending — it just starts over!"

✅ "My Plant Today" — What to Expect

- A bean planted in Lesson 2 (~7 days ago) should be at the **Sprout / Seedling** stage.
- Students may see the stem emerging, the cotyledons (first two thick leaves), and possibly the beginning of true leaves.
- Roots are usually not visible unless planted in a clear cup against the glass.
- **Praise all observations** — color, height, texture, direction of growth. There's no wrong answer for the observation drawing.

✅ Learning Objective — What Success Looks Like

- Child can name and sequence all four stages of the plant life cycle in order.
- Child can explain why it's a "cycle" (it goes around; seeds become plants that make more seeds).
- Child can identify what stage their own bean plant is at.
- Child can connect back to K-LS1-1: plants need sunlight, water, air, and soil/nutrients to survive.

🌟 **Bonus: Discussion Question Answers**

- **"Why a cycle, not a line?"** — Seeds → plants → more seeds. It circles back. A line has an end; a cycle doesn't.
- **"What if no seeds were made?"** — No new plants! The species would disappear. Seeds are survival.
- **"What does a seed need to start?"** — Water, warmth, soil. Then light + air for the growing plant.
- **"Do you have a life cycle?"** — Yes! Born → grow → have children → the next generation begins.

🎯 Ready to test your knowledge? → Take the Unit 2 Quiz!

10 fun questions about seeds, plant parts, sunlight, and the life cycle!

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